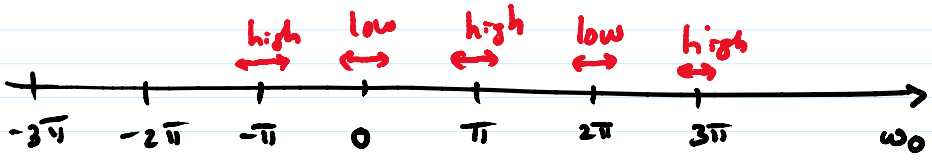
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DT Complex Exponential

support $(-\infty, \infty)$

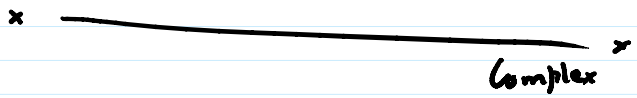
$$x(n) = e^{j\omega_0 n}, \quad n \in \mathbb{Z}$$

$$\omega_0 \in [-\pi, \pi] \quad \text{or} \quad [0, 2\pi] \qquad \omega_0 \equiv \omega_0 + 2\pi k, \quad k \in \mathbb{Z}$$



DT freq. variable.

(radians / sample)



- General form of DT Exponential

Supp. $(-\infty, \infty)$

$$x(n) = a^n ; \quad n \in \mathbb{Z}$$

$a \in \mathbb{C}$ complex number

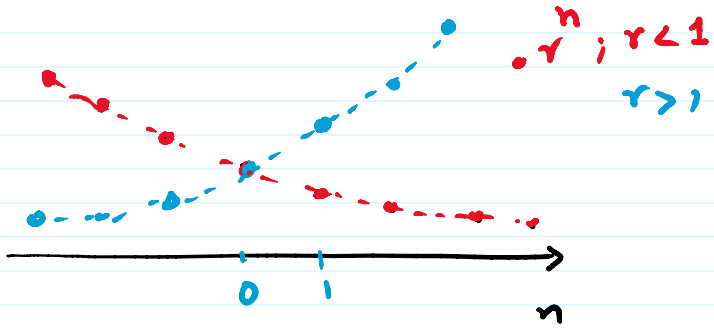
$$a = re^{j\omega_0}$$

Magnitude $|a| = r \geq 0$

Phase or angle $\angle a = \omega_0$

or
angle

$$x(n) = r^n e^{j\omega_0 n} ;$$



Properties of DT Signals

1. Periodic Signal : (*)

If $x(n) = x(n+N)$ for all n, | N integer

then $\{x(n)\}$ is periodic with Period N.

Smallest N which satisfies (*) is called fundamental period of Signal $\{x(n)\}$.

DT Complex Exponential $x(n) = e^{j\omega_0 n}$; $n \in \mathbb{Z}$

is periodic if & only if

$$\omega_0 = \pi \, p/q$$

p and q are integers.

$\omega_0 = \pi \, p/q$ p and q are integers.

Suppose N is period of $\{x(n)\}$

N is an integer $\quad x(n) = x(n+N)$ for all n

$$e^{j\omega_0 n} = e^{j\omega_0 (n+N)}$$

$$= e^{j\omega_0 n} \, e^{j\omega_0 N}$$

$$\Rightarrow e^{j\omega_0 N} = 1.$$

$$\Rightarrow \omega_0 N = 2\pi k \; ; \; k \in Z$$

$$\Rightarrow \boxed{\omega_0 = \frac{2\pi k}{N}}$$

To find fundamental period, find smallest positive integers k_1, N_1 such that

$$\omega_0 = \frac{2\pi k_1}{N_1}$$

N_1 is the fundamental period.

$$\omega_0 = \frac{2\pi \cdot 5}{100} = 2\pi \cdot \frac{1}{20}$$

x ——————— x

2. Symmetry :

Even Symmetry

$$x(n) = x(-n) \quad \text{for all } n$$

odd $\quad x(n) = -x(-n) \quad$ for all n.

Conjugate Even Symmetry.

$$x(n) = x^*(-n) \quad \text{for all } n$$

$(\,)^*$ → conjugation.

Conjugate odd Symm.

$$x(n) = -x^*(-n) \quad \text{for all } n.$$

Examples :

(1) $x(n) = \frac{1}{1+jn}$

(2) $x(n) = j\cos(\omega_0 n)$

$1+jn$

$$x(-n) = \frac{1}{1-jn}$$

conj. odd (verify)

$$x^*(-n) = \frac{1}{1+jn}$$

conj. even

---

3. Causal & anti causal Signals.

Causal signal :

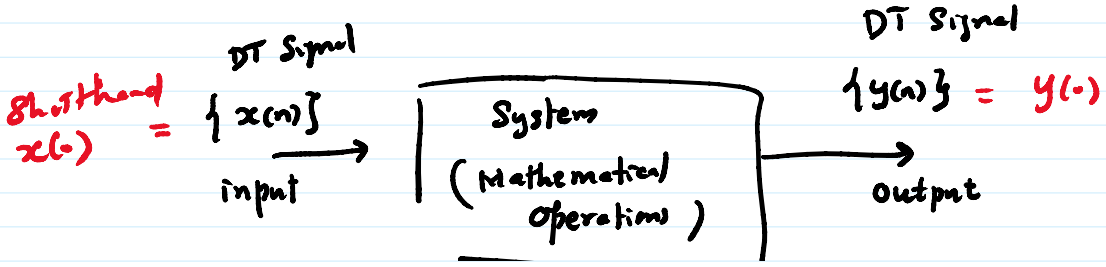
$x(n) = 0$ for all $n < 0$

Anti Causal

$x(n) = 0$ for all $n > 0$

Non Causal = NOT causal

Discrete Time Systems



input → | (Mathematical Operations) | → output

To specify system completely we need to specify

for all n, $y(n)$ = function $\{x(n)\}$

$y(n)$: output at time n

Entire input signal.

function may depend on time index n.

Examples:

① Identity System — stable, memoryless, causal

$$y(n) = x(n) \quad \text{for all } n.$$

② Amplifier — stable, memoryless, causal

$$y(n) = A\,x(n) \quad \text{for all } n$$

A = constant $\in \mathbb{C}$

③ Time Shift — stable, memory

n_0 is an integer. fixed.

causal if $n_0 \geq 0$

non causal if $n_0 < 0$

$$y(n) = x(n - n_0) \quad ; \quad \text{for all } n.$$

$y(n_0 - 1) = x(-1)$

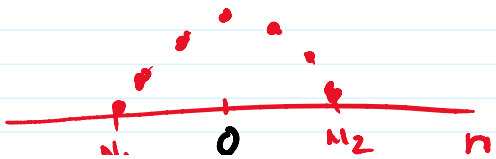
$y(n_0) = x(0)$

$y(n_0 + 1) = x(1), \ldots$

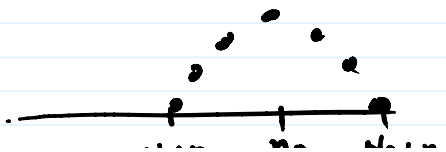
$n_0 > 0 \Rightarrow$ Delay.

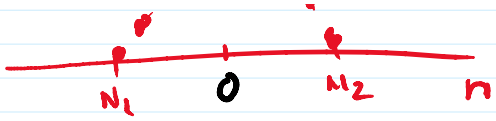
$n_0 < 0 \Rightarrow$ Advance

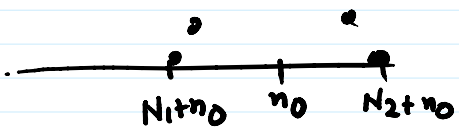
$\{x(n)\}$ Support $[N_1, N_2]$



$\{y(n)\}$ Support $[n_0 + N_1, n_0 + N_2]$







able

4. Time Reversal

Non Causal

$$y(n) = x(-n) \quad \text{for all } n.$$

$$\{x(n)\} = \{a, b, c, d, e, f\}$$

($n=0$ at b)

$$\{y(n)\} = \{f, e, d, c, b, a\}$$

($n=0$ at b)

stable

5. Time Reversal + Shift

$$y(n) = x(n_0 - n) \quad ; \quad \text{for all } n.$$

$$y(0) = x(n_0), \quad y(1) = x(n_0 - 1), \quad \ldots$$

$$\begin{cases} x_r(n) = x(-n) & ; \text{ Time reversal} \\ y(n) = x_r(n - n_0) & ; \text{ Time shift.} \end{cases}$$

$$= x(-(n - n_0))$$

$$= x(n_0 - n)$$

---

6. Accumulator.

"output at time n is the sum of all input samples upto time n".

$$y(n) = x(n) + x(n-1) + x(n-2) + \cdots$$

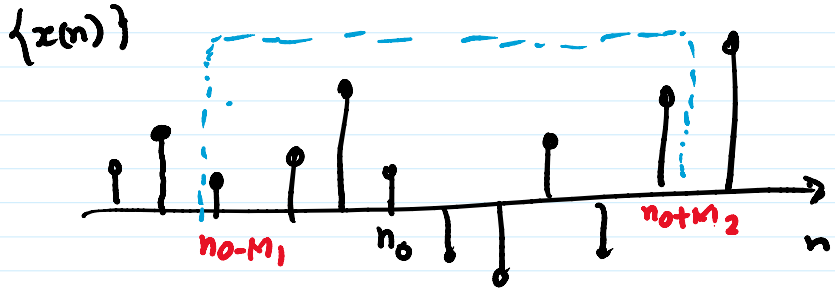
$$= \sum_{k=-\infty}^{n} x(k)$$

$\ell = n-k$

$$= \sum_{\ell=0}^{\infty} x(n-\ell)$$

stable

7. Moving Average filter



Non causal if $M_2 > 0$

$$y(n_0) = \frac{x(n_0-M_1) + \cdots + x(n_0) + \cdots + x(n_0+M_2)}{M_1+M_2+1}$$

$$y(n) = \frac{\sum_{k=-M_1}^{M_2} x(n+k)}{M_1+M_2+1}, \text{ for all } n.$$

$M_1 + M_2 + 1$

"Low Pass filter" will see later.

x ——————— x

8. Downsampling. Not Causal

M is an integer

$$y(n) = x(Mn) \quad ; \quad \text{for all } n.$$

$$y(0) = x(0)$$
$$y(1) = x(M)$$
$$y(2) = x(2M) \quad \cdots$$

$$\{y(n)\} = \{\cdots, x(-M), \underset{\uparrow}{x(0)}, x(M), x(2M), \cdots\}$$
$$n=0$$

DT System Properties

1. Memoryless

Output at any time n depends only at input sample at time n

y(n) depends only on x(n)

not past or future values of $\{x(n)\}$

2. Causal System

Output at any time index n

does NOT depend on

future input samples

op

$y(n)$ does not depend on $\{x(n+1), x(n+2), \ldots\}$

Causal $\quad y(n) = \cos(n+100)\, x(n-2)$

3. Stable System

A signal $\{x(n)\}$ is called bounded

if $|x(n)| \leq B$ (finite constant)

for all n

$0.5^n u(n)$ → Bounded : $\delta(n)$, $\sin(n)$, $\cos(\omega_0 n)$, $u(n)$, $100 e^{j\omega_0 n}$

Un Bounded $\quad x(n) = 2^n \; ; \; n \in \mathbb{Z}$

$x(n) \to \infty$ as $n \to \infty$

BIBO Stability

Bounded input Bounded Output

BIBO Stability

Bounded input Bounded Output

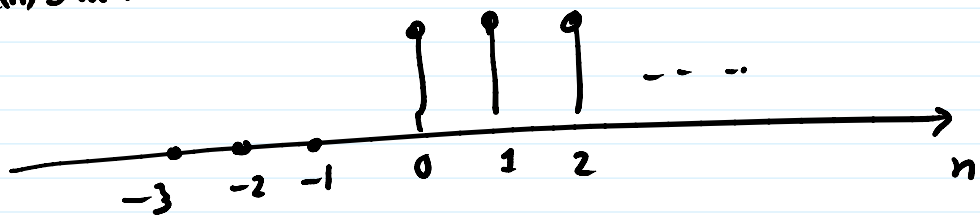
A system is BIBO Stable if the output remains bounded for any bounded input signal.

If there exists a bounded input for which system produces unbounded output then it is unstable system.

Accumulator (unstable)

$$y(n) = \sum_{k=-\infty}^{n} x(k)$$

Bounded $x(n) = u(n)$



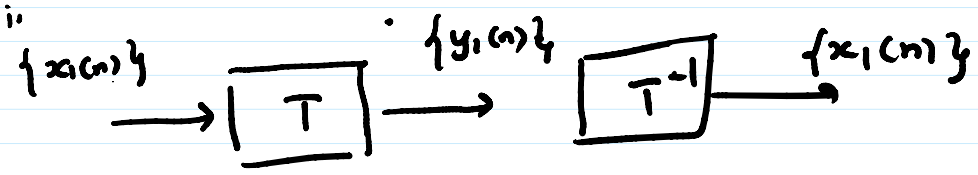
$$y(n) = \begin{cases} 0 & ; \quad n < 0 \\ n+1 & ; \quad n > 0 \end{cases}$$

$y(n) \to \infty$ as $n \to \infty$ (unbounded)

4. Invertible Systems.

Any Two distinct input signals always produce distinct output signals then System is invertible.

then system is invertible.



$$\{x_1(n)\} \xrightarrow{T} \{y_1(n)\}$$

$\{x_1(n)\} \neq \{x_2(n)\}$

$$\{x_2(n)\} \xrightarrow{T} \{y_2(n)\}$$

If T is invertible

$\{y_1(n)\} \neq \{y_2(n)\}$

If T is invertible then inverse system T^{-1} exists

such that

whenever $\{x(n)\} \xrightarrow{T} \{y(n)\}$

we have $\{y(n)\} \xrightarrow{T^{-1}} \{x(n)\}$